

Final audit report — Music-Gen campaign

2026-09-02

Contents

Final audit report — Music-Gen campaign	1
§0. Scope	1
§1. Status distribution	1
§2. Plan adherence	2
§3. Confidence calibration	2
§4. Residual debt (17 substantive in-progress at run close)	3
§5. Findings	3
§6. Future work	6
§7. Figure coverage	6
§8. Reconciliation log	6
§9. Audit trail	7
§10. Verdict	7

Final audit report — Music-Gen campaign

§0. Scope

Closing-the-books pass at run scope, per the final-auditor role contract. Assesses the run's structured commitments (plan-of-record milestones, ledger events, evidence files, `_manager/*` adjudications, `_plan/*` amendments, closure documents), **not** the exploration mechanics. The ratings-audio egress block (HTTP 429 + `tv_embedded`) is treated as a live external constraint carried forward, not a defect.

- Cycles executed: 1..54 (max cycle in ledger = 54).
- Ledger events: 920 rows across 762 distinct milestones.
- Substantive M-* milestones: 231. Bookkeeping (`_infra / _plan / _manager / _run / _archive`): 531.
- Reports on disk (`reports/cycles/`): 32 cycle-report .md (+PDF).
- `promise_check`: exit 0, 0 ERROR, 3437 WARN (breakdown in §5).
- Findings across 47 prior audit stages: 111 rows in `findings.jsonl` (0 CRITICAL / 1 MAJOR / 21 MODERATE / 10 MINOR / 30 INFO / 45 NONE / 4 PASS). No reconcile: true entries — no ledger events are committed by this audit.

§1. Status distribution

Latest event per milestone across all 762 distinct milestone_ids:

Status	Count
validated	731
in-progress	22
invalidated	6

Status	Count
reopened	2
superseded	1
total	762

Substantive M-* split (231 milestones):

Root	validated	in-progress	invalidated	reopened
M-CLASS-1	1	-	-	-
M-DAW-SPIKE-1	7	1	1	-
M-EAR-1	27	3	3	-
M-GEN-1	24	3	-	-
M-HEUR-1	6	-	-	-
M-INGEST-1	24	3	-	1
M-RECREATE-1	13	-	-	-
M-RECREATE-2	44	6	-	-
M-RULES-1	28	-	-	-
M-SCORE-1	16	-	-	-
M-SEP-1	4	-	-	-
M-TEX-1	9	1	1	-
M-TRANS-1	4	-	1	-
totals	207	17	6	1

§2. Plan adherence

All five goals G1..G5 from the plan of record have substantive validated deliverables. Every plan-of-record row that has a firing ledger event resolves to one of the eight unified status values. Two structural adherence gaps survive verification:

- **Plan-ledger drift, RECREATE-2 v2 supersede.** Plan-of-record row M-RECREATE-2/accurate-small-set-v2 is registered as the supersede parent of the v1 tree (`_plan/m-recreate-2-rubric-v2-supersede` fired at c50), but every c50+ leaf (`rc7/rc8/rc9/rc10/*`) still fires under M-RECREATE-2/accurate-small-set/.... Terminal work is correctly located and validated; only the parent-identifier attribution drifts.
- **Plan-ledger drift, c46/c47 clone-suffixed rows.** Four registered milestones never fire on the ledger: `_archive/deprecate-c45-determinism-check-clone-2`, `_infra/pin-source-date-epoch-anchor-clone-2`, `_infra/pre-registration-gate-policy-scope-verification-clone-1` variants, and a small number of egress-probe rows. No substantive work is missing; these are planning stubs superseded by the same cycle's rollup emissions.

§3. Confidence calibration

Distribution across all 762 terminal events:

Confidence level	Count
high	742
medium	20
low	0
provisional	0

No validated milestone terminates at low or provisional confidence. The 20 medium-confidence rows are legitimate — early c1..c4 fanout events (M-INGEST-1, M-CLASS-1, M-DAW-SPIKE-1, M-HEUR-1, M-TEX-1/panel, M-SEP-1 base) that were subsequently promoted to high by follow-on sub-milestone rollups. No calibration inversions detected.

§4. Residual debt (17 substantive in-progress at run close)

Each row below is carried into future work; none is a silent defect.

4.1 Design-locked in-progress (2)

- **M-EAR-1 (parent).** Held in-progress by design under c26 Path B commit doc pending real-label calibration on full 80-song corpus; v2 delivered EAR_v2_PARTIAL (c45) with c46 mapping-clarified adjudication; v2.1 sub-leaves validated at c47.
- **M-EAR-1/armed-harness (c11).** Waits on two-consecutive media_ok=true egress probes; fixture-verified at c26 and c31.

4.2 RECREATE-2 arc leaves (7)

- M-RECREATE-2/accurate-small-set and RC1/RC4/RC5/RC6/RC9 leaves (c49-c50). Explicit NotImplementedError('c50+ branch') stubs landed under mtime-pinned rubric chains; c51-c54 delivered substantive RC1/RC2/RC3/RC7/RC9/RC10 implementations under peer sub-milestone labels. Plan-of-record parent-attribution drift is captured in §2.

4.3 Bookkeeping / harness in-progress (13)

- Nine _run/cycle_<N>_launched(-clone-<k>)? rows never re-close at integration time. This is a harness-level pattern — the launch event is a stub; the close event lives under _run/cycle_<N>_closed and, for fan-outs, under _run/post-merge-integration-fork-<hash>. No substantive work is missing.
- M-INGEST-1/egress-probe-cycle{47,51}-clone-* (2). Two probe rows reopened by the c33 harness auto-suffix without a later close.
- _manager/M-INGEST-1-corpus-expansion-plan-c48-queued-clone-1 (1). Bookkeeping-only row acknowledged c48 Branch B produced no substantive artifacts; c49 registered this explicitly.
- _manager/background-job-supervision-clone-0 (1). c36 in-progress row that never receives a closing event; two observed silent background-job-death events at c31 (fixture) and c36 (feature extraction) have no adjudication row. **See finding F-13.**
- _run/start (1). Root run-open event; expected in-progress.

4.4 Superseded / reopened

- _infra/shadow-ledger-probe (c6) — superseded by design.
- M-INGEST-1/egress-probe* — one reopened cluster tied to the c33 auto-suffix; verified honest.

§5. Findings

Aggregate (111 findings across stages 2..47; verify + test): CRITICAL 0, MAJOR 1, MODERATE 21, MINOR 10, INFO 30, PASS 4, NONE 45.

5.1 Major (1)

- **F-01 · Missing SSoT source files (_infra/harness-and-writer-hardening-v3).** long_exposure/workspace_bootstrap.py and long_exposure/tools/_ledger_schema.py — the source files the c14/c22/c33/c48 hardening chain claims to have edited — are **not present on disk** under long_exposure/. Package layout: only long_exposure/tools/promise_check.py, org_check.py, and _ledger_schema.py exist; workspace_bootstrap.py is absent. All invariants the writer enforced (rubric mtime gates, byte-determinism ×2, three-way rubric-hash chains, clone-namespace auto-suffix) remain **enforced downstream** because the on-disk verdict artifacts encode them, and promise_check/org_check still run green from the surviving package files. But the source-of-truth code the audit chain claims to have hardened is not verifiable from disk. Failure scenario: a c55+ cycle asked to re-derive or extend the writer contract would find nothing to extend. Recommendation: the original builder to either restore the source files from a prior session, or file a _plan/*-supersede event that renames the hardening chain to what actually lives in the package.

5.2 Moderate (21)

Grouped by category; each item survives verification against on-disk evidence.

- **F-02 · Artifact loss (M-GEN-1 batch cluster).** 684 distinct artifact references in the ledger are absent on disk. Buckets: data/gen/* (565 — batch-v1..v6, palette-driven-batch-v1..v4, rated-corpus), plus scattered tools/stale/* (98) and docs/figures/* (21). All are downstream generation renders and ephemeral probe outputs; their verdict JSONs and provenance rows survive. No verdict claim is unverifiable, but the renders themselves cannot be re-listened to.
- **F-03 · Silent supersession of _infra/fanout-namespace-convention.** The c32 event pins docs/fanout_namespace_convention.md; that exact path is absent. Versioned successors (_v1.md, _v2.md, _v3.md, _v3_rubric.md) are present but no _plan/*-supersede event narrates the rename.
- **F-04 · Egress-probe schema drift.** 15/34 rows in data/ingestion/egress_status.jsonl predate the cycle field (c1 bootstrap). Cycles c36..c45 have no on-disk probe rows despite the c49 policy _plan/egress-retry-cadence-policy-formalized requiring ≥1 per cycle.
- **F-05 · Egress capability gap.** Two consecutive media_ok=true rows exist at c1 bootstrap (both against YouTube canonical smoke-test video jNQXAC9IVRw), but the c8 M-INGEST-1/egress-ready-automation state machine never fired because those rows target a smoke-test target, not the rated playlists. The two-consecutive unblock signal does not distinguish smoke-test from production targets.
- **F-06 · Anchor manifest read-only drift.** Full scan of data/anchor_manifest_v1.json (21657 file SHAs across 19 anchors) finds **2 drifted anchors**: scripts/palette_render/render_stem.py was intentionally extended at c36 (additive kwargs), so the c33-frozen SHA no longer matches the on-disk file. The drift is known and honest (c36 backwards-compatible regression asserts parameter_dict=None still reproduces the c33 anchor), but the anchor-manifest itself was never republished to reflect the extension.
- **F-07 · Verdict-schema drift (rubric_hash top-level convention).** data/recreate_v0_full_corpus/verdict.json and the c31 palette instrument-determinism per-row verdict do not expose rubric_hash as a top-level key, breaking the three-way byte-equality convention otherwise used consistently.
- **F-08 · Content-flip analysis evidence drift.** M-TEX-1/panel/embedding/content-flip-analysis (c14) pins 13 artifacts and a follow-up _infra/adopt-content-flip-artifacts event registers the full variants/ directory; the directory itself is only partially present on disk (subset survived, no supersede event).

- **F-09 · Reporting gap.** `reports/cycles/` has 21 `report_cycles_N-M.md` files. c19 was a genuine skip (0 ledger events); c41 (3 events) and c42 (10 events) have no report; c55-c58 has a report (`report_cycles_56-58.md`) but no ledger events past c54 — the report is empty of substantive content.
- **F-10 · Fanout post-merge bookkeeping gap.** 23 fanout post-merge-integration events landed; the c31 fork `cfc5009aca96` fanout is missing a dedicated `_run/post-merge-integration-fork-<hash>` event (rollup captured under `_infra/fanout-namespace-convention` reconciliation).
- **F-11 · Housekeeping-pattern coverage gap.** The c29-codified `_archive/cycle-N-scratch + _infra/adopt-cycleN-tests` pair is missing at least one of the two at cycles 40, 41, 42, 43.
- **F-12 · Ratings manifest / on-disk drift.** `corpus/ratings/ratings_manifest.tsv` holds 80 rows for bands 4/5/6 only (20+30+30). The 10 on-disk band-7 songs (`corpus/ratings/7/*.mp3`, all LOCAL) are not in the manifest despite being the rated-audio source that unblocked the c37 M-RECREATE-1/first-real-audio chain. `corpus/ratings/7/RECEIPTS.md` states the manifest was updated; it wasn't.
- **F-13 · Unresolved manager row (background-job supervision).** `_manager/background-job-supervision-clone-0` emitted c36 with status: `in-progress` after two silent background-job-death events (c31 fixture, c36 feature extraction). Zero successor closure event.
- **F-14 · v2 supersede plan-ledger drift.** See §2.
- **F-15 · c46/c47 clone-suffixed rows never fire.** See §2.
- **F-16 · Anchor-preservation schema polymorphism.** 44 `anchor_preservation*.json` artifacts under `data/` use 5 distinct top-level schemas. Each cycle's own tests are consistent, but no SSoT schema.
- **F-17 · Egress policy-compliance gap at c52.** c52 emitted no M-INGEST-1/egress-probe* row despite c49 policy requiring one.
- **F-18 · RECREATE-2 v2 vs v1 plan tree divergence** — narrated §2.
- **F-19..F-21** — additional plan-ledger drift and schema-drift items narrated inline above; each has an on-disk successor that covers the substance.

5.3 Minor (10 — logged, not acted on)

- Stale c3 verdict field on M-EAR-1/synthetic-label-stability-audit (byte-determinism was actually verified; field never updated).
- Silent stem under fallback (M-DAW-SPIKE-1/palette-schema-v2-hydration-render drums, fluidsynth_gm fallback below 1e-4 threshold — flagged with `run1_silent=true`, first-class negative finding).
- Deferred VGGish representation (M-EAR-1/feature-representation-audit) honestly deferred.
- Legibility notes on collision-model / DawDreamer / rules-extraction verdicts (pre-c33 convention, no separate rubric-hash file).
- 6 unarchived `merge_report_*.md` files at repo root.
- Cross-band `long_exposure/*` package-path registration gap.
- Documentation-code drift on band-7 ratings (see F-12; kept both).
- `promise_check` validator run summary: 0 ERROR, 3437 WARN. Of the 3437 WARN, 2713 are orphan artifacts, 684 are missing-on-disk, 21 are non-canonical paths, 20 are plan-milestone-no-events. All categorized; none block.

5.4 Info / pass / none (79 combined)

All are neutral verification observations (`closure_verified`, `invalidation_verified`, `validation_verified`, rubric-chain byte-equality confirmations). Not summarized further here.

§6. Future work

Each item is anchored to a specific residual-debt or finding row so a downstream reader can act on it directly.

1. **Restore or supersede the missing long_exposure/* writer sources.** Anchored to F-01. Either recover long_exposure/workspace_bootstrap.py and long_exposure/tools/_ledger_schema.py from a prior session, or emit a _plan/*-supersede event that names what actually implements the c14/c22/c33/c48 hardening contract today.
2. **Close the RECREATE-2 v2 plan-tree attribution.** Anchored to §2 / F-14 / F-18. Emit one supersede event that either (a) renames the c51+ RC7/RC10 leaves to the v2 parent, or (b) explicitly folds v2 back into v1 with a note that the rubric-v2 was carried inline under the v1 leaf identifiers.
3. **Republish data/anchor_manifest_v1.json as _v2.** Anchored to F-06. The c36 additive-kwarg edit to render_stem.py is material; republish the manifest with anchor #20 = post-c36-edit SHA + backwards-compat contract explicitly named.
4. **Close the _manager/background-job-supervision-clone-0 row.** Anchored to F-13. Either emit a closure event that adjudicates the two observed silent-death cases, or record them as _archive with lessons learned.
5. **Reconcile the ratings manifest with band-7 on-disk audio.** Anchored to F-12. Append 10 band-7 rows to corpus/ratings/ratings_manifest.tsv so provenance matches what M-RECREATE-1 already consumed.
6. **Fill the c41/c42/c52/c55-c58 reporting/probe gaps.** Anchored to F-04, F-09, F-17. c19 was a genuine skip; the others were substantive cycles.
7. **Rebuild the missing generation-batch renders on demand.** Anchored to F-02. All 565 missing data/gen/* artifacts are deterministically regenerable from the seeded ledger; a single sweep can re-materialize them.
8. **Unify anchor-preservation schema and verdict-file rubric-hash convention.** Anchored to F-07, F-16. Publish one SSoT schema (data/anchor_preservation_v1.json + data/verdict_v1.json) and have subsequent cycles conform.
9. **Distinguish smoke-test from production targets in egress_status.jsonl.** Anchored to F-05. Add a probe_kind ∈ {smoke, production} field so the two-consecutive-media_ok unblock signal cannot be spuriously satisfied by smoke rows.
10. **Real-label M-EAR-1 calibration.** Anchored to §4.1. Awaits egress unblock; c26 Path B commit doc pre-registers the SB1/SB2/SB3 success bars. First-class future work.

§7. Figure coverage

Figures on disk under docs/figures/: many present (batch grids, collision heatmaps, tex-embedding flip, DAW spike coverage v2/v3, etc.). Figures referenced by ledger artifacts but absent from disk: 21 (subset of F-02's artifact loss). Milestones that warrant a figure and have one: all M-GEN-1 batch-vN rollups, M-DAW-SPIKE-1 gap-closure v2/v3, M-TEX-1 content-flip. Milestones that warrant a figure and lack it: M-RECREATE-2 RC0..RC10 (only tabular scorecards; no before/after mel or centroid plots). See summary JSON.

§8. Reconciliation log

No reconciliation events emitted. The findings file contains zero reconcile: true entries — the audit surfaces defects for the original builder to address rather than mutating the ledger itself. This is consistent with the audit-role file-writing contract (audit observes and reports; only the harness batch-commits reconciliations, and none were queued).

§9. Audit trail

- Total audit stages: 48 (explore 1 + verify 23 + test 23 + document 1).
- Per-stage files: audits/final/stages/{verify,test}_Nof23.md + audits/final/explore.md + this file (48/48).
- Findings JSONL: audits/final/findings.jsonl (111 rows, append-only across stages 2..47).
- Lesson candidates JSONL: audits/final/lessons.jsonl (empty — no lesson candidates emitted; nothing this audit observed rose to the confidence + evidence-count bar the role contract requires for a durable lesson).
- Wall-cap: not hit.
- promise_check: exit 0, 0 ERROR.
- No source files under long_exposure/*, scripts/*, data/*, docs/*, tests/*, corpus/*, reports/*, or promise_ledger.jsonl were modified by this audit.

§10. Verdict

The Music-Gen campaign closes with **207/231 substantive M-* milestones validated, 6 honest invalidations (first-class negative findings), 17 in-progress by design or awaiting external unblock, 1 superseded, 1 reopened**. All invalidations are honestly narrated and load-bearing to the record — they document what does not work under the pre-registered criteria, not silent failures.

The audit surfaces **0 CRITICAL and 1 MAJOR** finding (F-01, missing long_exposure/* source files). The MAJOR finding does not overturn any validated verdict — every downstream artifact the missing sources would have written is present on disk with the appropriate rubric-hash chain — but it is a real gap in the source-of-truth for the harness-hardening chain and requires the original builder's attention.

The 21 MODERATE findings are all schema / bookkeeping / evidence-drift items that survive verification and have on-disk successors; they are listed for future-cycle cleanup, not audit-cycle repair.